

RESEARCH ARTICLE

Improving the piezoelectric properties of CBN-based ceramic by a Ce ion

Ning Chen | Fei Wang | Xu Li | Yining Xie | Jingwen Xi  | Shangyi Guan  | Wei Shi | Hao Chen | Jie Xing  | Jianguo Zhu 

College of Materials Science and Engineering, Sichuan University, Chengdu, China

Correspondence

Jie Xing and Jianguo Zhu, College of Materials Science and Engineering, Sichuan University, 610064 Chengdu, China.

Email: xingjie@scu.edu.cn and nic0400@scu.edu.cn

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Abstract

CaBi₂Nb₂O₉ (CBN), one of the bismuth-layered structural ferroelectrics, with high Curie temperature (T_C), has great potential in high-temperature applications. In this work, high Curie temperature and piezoelectric constant (d_{33}) are realized in modified CaBi₂Nb₂O₉ ceramics with Ce-substitution. Ce-substitution changes the crystal structures and domain structures of CBN-based ceramics, so as to improve the piezoelectric properties. The optimal performances are obtained with a high d_{33} value (~ 18.0 pC/N) and a T_C value ($\sim 930^\circ\text{C}$), together with a low $\tan \delta$ value (~ 0.028 at 500°C). Moreover, the thermal stability is also enhanced, where the d_{33} value maintains 93.9% of its original value after annealing at 900°C for 2 h. Thus, these findings play a meaningful role in devices manufacturing, where the apply temperature is often more than 500°C .

KEYWORDS

CaBi₂Nb₂O₉, doping, lattice distortion, oxygen vacancies, piezoelectric constant

1 | INTRODUCTION

Piezoelectric acceleration sensors, as function of vibration test calibration, mechanical dynamic experiment, environmental simulation test, and fault diagnosis, are widely employed in nuclear power industry, oil production, aerospace, and so forth.¹ As is known, the working temperatures in these fields usually exceed 600°C , some of which even reach 1000°C , with more strict demands for the performances of sensors.^{1,2} From the perspective of applications, the materials for sensors ought to have following characteristics: (1) good piezoelectric properties to guarantee sensitivity of sensors, (2) high T_C values to ensure that the service temperature of piezoelectric materials is high enough, which is always lower than half of Curie temperature accounting for the depolarization effect of ferroelectrics, (3) stable piezoelectric performance and large resistivity at high temperature

to let engines run steadily for long time under a high-temperature condition. Therefore, the piezoelectric materials with T_C values higher than 800°C can fit the requirements.

As a result, bismuth-layered structural ferroelectrics (BLSFs), with the formula $(\text{Bi}_2\text{O}_2)^{2+}(\text{A}_{m-1}\text{B}_m\text{O}_{3m+1})^{2-}$, have aroused significant attention due to their low dielectric loss values and high T_C values (which are usually above 500°C even surpassing 900°C). The BLSFs consist of perovskite-like slabs $(\text{A}_{m-1}\text{B}_m\text{O}_{3m+1})^{2-}$ and bismuth oxide $(\text{Bi}_2\text{O}_2)^{2+}$ layers, where m is the number of perovskite layers between the Bi_2O_2 layers varying from 1 to 6. A-site in the perovskite layers is occupied by ions with low valence and large radius that suit dodecahedral coordination or their compounds (including K^+ , Ce^{3+} , Pr^{3+} , La^{3+} , $[\text{K}_{0.5}\text{Ce}_{0.5}]^{2+}$, and $[\text{Li}_{0.5}\text{Ce}_{0.5}\text{Pr}_{0.5}]^{2+}$), whereas B-site usually contain transition ions or their compounds with a relatively small radius, high valence (such as W^{6+} ,

Ta^{5+} , Mo^{6+} , Nb^{5+} , and $(\text{Cu}_{0.5}\text{Ta}_{0.5})^{4+}$.³ Among them, CBN ceramics ($m = 2$) with highest T_C values (about 940°C), have enormous potential for high-temperature industries.⁴ Nevertheless, the piezoelectric constant of pure CBN ceramic is ~ 6.0 pC/N,⁵ which heavily limits the sensitivity of devices. Hence, it is necessary to improve piezoelectric properties of BLSFs to satisfy devices applications.

As is known, the d_{33} value of BLSFs is poor because the spontaneous polarization is confined to the a - b plane other than the vertical direction. Both chemical doping and processing improvement can lead to structural distortion and thus improve the piezoelectric constant. For example, Liu et al. prepared $\text{Ca}_{0.92}(\text{Na}_{0.5}\text{Ce}_{0.5})_{0.08}\text{Bi}_2\text{Nb}_2\text{O}_9$ ceramics, demonstrating a high d_{33} value of 16.1 pC/N and good thermal stability,⁶ with a high d_{33} value (~ 17.6 pC/N) and a relative low T_C value ($\sim 904^\circ\text{C}$) in Na/Bi/Li/Ce-doped CBN ceramics.⁷ In addition, B-site co-substitution CBN ceramics such as MoCr, WCr, and WCu also exhibited improved d_{33} values to 15.0, 15.0, and 14.0 pC/N, respectively.^{8–10} Among them, the substitution of A-sites is a feasible way to improve piezoelectric properties of CBN-based ceramics. However, how to simultaneously achieve good piezoelectric properties and a high T_C value of CBN-based ceramic is desired to devote efforts at present.

In general, a d_{33} value is negatively correlated with a T_C value in other reports.^{6,8,9} Previously, it was reported that the CBN-based ceramic exhibits a high T_C value ($\sim 947^\circ\text{C}$) and excellent resistance performance ($\sim 1.53 \times 10^6 \Omega \text{ cm}$ at 600°C) but a poor d_{33} value (~ 13.0 pC/N) by introducing Gd ion at A-site.¹¹ Based on (Li/Ce, X) co-doping at A-site, CBN-based ceramics show high d_{33} values of 17.2, 17.3, and 17.0 pC/N for $X = \text{Bi, Pr, Y}$,^{5,12,13} respectively, indicating the potentiality of improving d_{33} values via Li/Ce ions. Therefore, Ce, Li, and Gd ions are adopted to improve the performances of CBN-based ceramic. In this work, $\text{Ca}_{0.86}(\text{LiGd}_{1-x}\text{Ce}_x)_{0.07}\text{Bi}_2\text{Nb}_2\text{O}_9$ ceramics are prepared by a traditional solid-state method. Their structures, electrical properties, piezoelectric properties, and so forth are studied. Finally, the ceramics with the formula $\text{Ca}_{0.86}(\text{LiGd}_{0.6}\text{Ce}_{0.4})_{0.07}\text{Bi}_2\text{Nb}_2\text{O}_9$ exhibits enhanced piezoelectric properties ($d_{33} \sim 18.0$ pC/N), high T_C values, and good thermal stability maintaining 93.8% of its original value after annealing at 900°C for 2 h simultaneously.

2 | EXPERIMENTAL PROCEDURES

The experiment and characterization of $\text{Ca}_{0.86}(\text{LiGd}_{1-x}\text{Ce}_x)_{0.07}\text{Bi}_2\text{Nb}_2\text{O}_9$ ceramics (CBNLG-C100x, $x = 0.0, 0.2, 0.4, 0.6$, and 0.8) are given in the [Supporting Information](#).

3 | RESULTS AND DISCUSSIONS

As shown in Figure 1A,B, better performances that enhance d_{33} values, high T_C values, and better thermal stability are simultaneously obtained in CBNLG-C100x ceramics, indicating its high potentiality in high temperature applications. The optimum d_{33} value of 18.0 pC/N is achieved in CBNLG-40C ceramic, which is two times larger than that of the non-doped ceramic (8.6 pC/N), indicating that Ce ion can effectively improve the piezoelectric properties. The increase of d_{33} values may be resulted from intrinsic and extrinsic factors. For intrinsic aspect, Ce-doping causes crystal distortion occurring, which changes crystal symmetry and contributes to the enhancement of piezoelectric properties. For extrinsic aspect, the domain structure is closely related to piezoelectric properties. Comparing CBNLG-0C ceramics with CBNLG-40C ceramics, the size of domains decreases, which is beneficial to an easy reversal of domains and for further enhancement of piezoelectric activities. In conclusion, piezoelectric properties can be improved via the doping of Ce. Meanwhile, the d_{33} values of all samples are still higher than 88% of their initial values after being annealed at 900°C for 2 h. Especially, the d_{33} values of CBNLG-20C and CBNLG-40C ceramics maintain, respectively, 93.7% and 93.9% (Figure S1a) of their initial values, suggesting that Ce-doping can improve thermal stability. The d_{33} values do not become zero when the annealing temperature exceeds T_C . Because the pinning effect of domain walls as a consequence of produced oxygen vacancies leads to the incomplete switching of domains during the annealing process.¹⁴ The T_C values of CBNLG-C100x samples are 945, 932, 930, 926, and 928°C respectively, the relationship of which is similar to that observed between a/b and x , which is caused by the symmetry of a crystal structure.¹⁵ In general, the T_C value is closely related to the symmetry of a crystal structure, which can be represented by an a/b ratio given from the refinement results.^{15–17} The larger a/b value corresponds to the lower symmetry of the crystal structure. Moreover, the T_C value decreases with increasing symmetry. As shown in Figure S6c, with increasing Ce ion, the gradual decrease of a/b values of CBNLG-C100x ($x \leq 0.8$) means that the crystal symmetry is enhanced, leading a lower T_C value. When the Ce ion further increases, the crystal symmetry is weakened, thus leading to a slight increase of T_C . Compared with different doping elements in CBN-based ceramics, this work exhibits better comprehensive properties as shown in Figure 1C.^{6–10,12,18–21}

To further evaluate the origin of improved piezoelectric properties, the X-ray powder diffraction (XRD) patterns of CBNLG-C100x ceramics with 2θ ranging from 20 to 60° are shown in Figure 2A₁. No second phase is observed, and all patterns can be indexed by JCPDF card 49-0608,

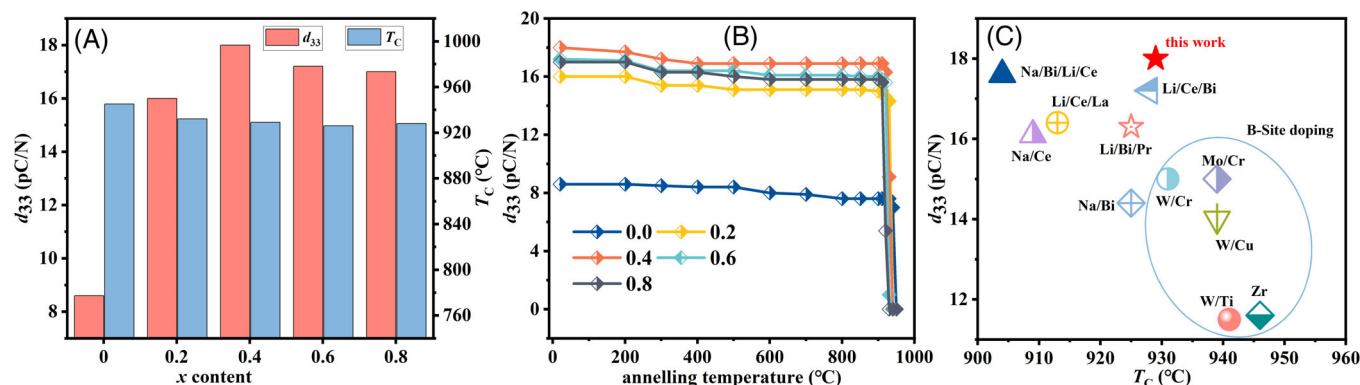


FIGURE 1 (A) The d_{33} values measured at room temperature and T_C values of CBNLG-C100x ceramics; (B) thermal stability of CBNLG-C100x ceramics; (C) d_{33} values versus T_C values of CBN-based ceramics with different doping elements

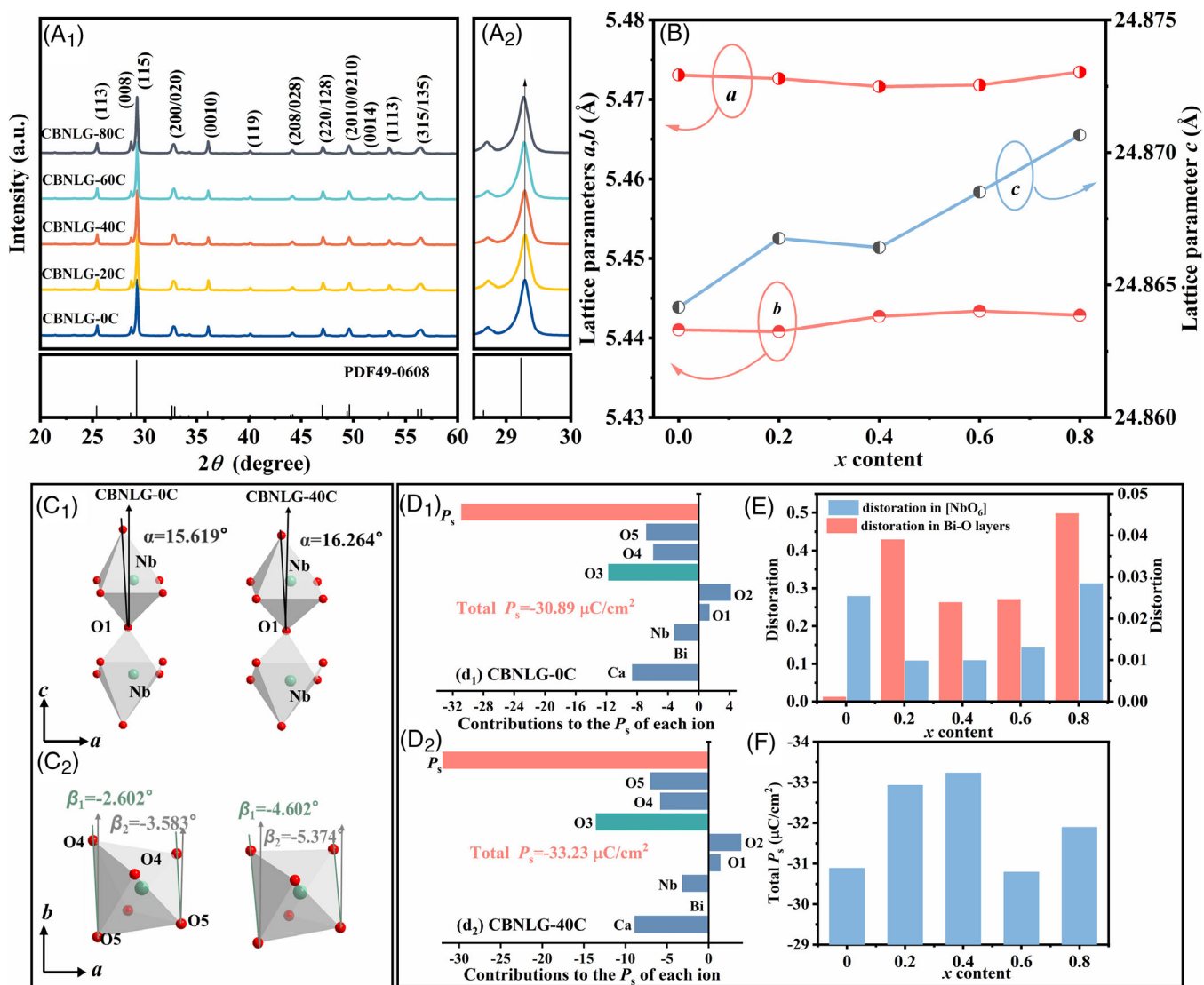


FIGURE 2 (A₁) X-ray diffraction (XRD) patterns and (A₂) enlarged peak (115) patterns of CBNLG-C100x ceramics; (B) the lattice parameters a , b , and c as a function of x content; schematic representations of (C₁) octahedral tilt from the c axis, and (C₂) octahedral rotation in the a - b plane for CBNLG-C100x ($x = 0.0, 0.4$); the contribution to the P_s of each ion of (D₁) CBNLG-C0 and (D₂) CBNLG-C40 ceramics; (E) the distortion in Bi-O layers and $[\text{NbO}_6]$ octahedrons; (F) total P_s value as a function of x content

demonstrating that the doped cations have diffused into lattice and formed a stable solid solution. The intensity of a (115) diffraction peak of all samples is the strongest, which is in conformity with the highest peak of BLSFs (112 $m+1$).²² From Figure 2A₂, the peak (115) slightly shifts to a lower angle after the introduction of Ce, indicating the increase of lattice constants. The increase of lattice constants can be attributed to the instability of cerium ions, resulting in the coexistence of Ce³⁺ and Ce⁴⁺, and more details are in the X-ray photoelectron spectroscopy (XPS) chapter. Moreover, Ce³⁺ replaces Gd³⁺, whereas Ce⁴⁺ replaces Nb⁵⁺. On the basis of the law concluded by Shannon, the ion radius of Gd³⁺ (CN = 12) is smaller than Sm³⁺ (CN = 12, ~1.24 Å).²³ Hence, this trend in diffraction peaks can be ascribed to the smaller ion radius of Gd³⁺ (<1.24 Å, CN = 12) than Ce³⁺ (CN = 12, ~1.34 Å) and Nb⁵⁺ (CN = 6, ~0.64 Å) than Ce⁴⁺ (CN = 6, ~0.87 Å).²³ To further study the influence of Ce-doping on a crystal structure, the Rietveld method is employed to refine the powder XRD data of samples (shown in Figure S2a–e) with an orthorhombic space group *A21am*. From Figure S2, the refinement results are in good agreement with the experimental ones, confirming the reliability of refinement data. The lattice parameters refined are presented in Figure 2B, showing no obvious change with parameters *a* and *b*. The lattice constant *c* increases slightly with the increasing of *x* content.

To investigate crystal structure distortion more intuitively, visualized crystal structures of CBNLG-C0 and CBNLG-C40 ceramics are obtained based on atomic positions from the XRD Rietveld refinements (Figure 2C₁, C₂), whereas the crystal structures of rest ceramics are shown in Figure S3a₁, a₂. Figure 2C₁ is viewed from the direction of (010) and Figure 2C₂ is observed from the (001) direction. It is known that the substitution will cause the distortion of NbO₆ octahedrons between perovskite-like layers. From Figures 2C₁ and S3a₁, the tilt angles along the *c* axis of NbO₆ octahedrons in Ce-doped CBNLG ceramics, which can be represented by the bond angle ∠O₂–O₁–*c* (*a* = 15.797, 16.264, 16.474, 15.719°, respectively), are bigger than CBNLG-C0 ceramic (15.619°), suggesting that the Ce ion increases the tilt of NbO₆ octahedrons. More particularly, the *a_x* and *a_y*, representing the tilting angles of NbO₆ octahedrons at *a*–*c* plane and *b*–*c* plane, respectively, are given in Figure S3b. Both *a_x* and *a_y* are negative, and the value of *a_y* is closed to *a*. In bismuth-layered structure ferroelectrics, the existence of the slip plane perpendicular to the *b* axis and mirror planes along the *c* axis can lead the direction of spontaneous polarization parallel to the *a* axis, and thus the ion displacement along the *b* and *c* axes has no impact on *P_s* values.²⁴ As exhibited in Figure S3b, the *a_x* varies within a very small range and maintains a smaller value (approaching to –3°) compared to *a_y*

(~–15°), indicating that the tilt of NbO₆ octahedron plays a tiny role in spontaneous polarization. There are rotations along the *b* axis in the *a*–*b* plane of NbO₆ octahedron beside the tilt, which refers to the angles β₁ and β₂ in Figure 2C₂. As shown in Figure S3c, the angles β₁ and β₂ have the same variation tendency and reach the maximum (β₁ = –4.602°, β₂ = –5.374°) when the doping content equals 0.4. The crystal distortion is closely related to the *P_s* of ceramics. In order to describe the overall distortion caused by doping into the oxygen octahedrons and bismuth oxygen layer, the square deviation of bond length given in Table S1 is defined as the lattice distortion.²⁵ Viewed from Figure 2E, the distortion in Bi–O layers is higher than that in NbO₆ octahedrons, which makes main contribution to *P_s* values. Furthermore, it suggests that the Ce-doping gives rise to lattice distortion. At the same time, the *P_s* value of CBNLG-C100*x* (*x* = 0–0.8) ceramics can be calculated by Shimakawa's mode²⁶:

$$P_s = \sum_i \frac{m_i \times \Delta x_i \times Q_{ie}}{V}, \quad (1)$$

where *m_i* is site multiplicity, Δ*x_i* is atomic displacement along the *a* axis from corresponding position in tetragonal structure, *Q_{ie}* is the ionic charge of the *i*th constituent ion, and *V* is the volume of unit cell.²⁷ The *P_s* values of CBNLG-C0 and CBNLG-C40 ceramics are shown in Figure 2D₁, D₂, whereas the *P_s* values of CBNLG-C100*x* (*x* = 0.2, 0.6, and 0.8) ceramics are given in Figure S3d₁–d₃; the values are 30.89, 32.93, 33.23, 30.93, and 31.89 μC/cm², respectively. For each composition, the total spontaneous polarization mainly comes from Ca, O₄, O₅, and O₃ ions. Furthermore, the calculated *P_s* values increase initially and decrease afterward, and the sample with *x* = 0.4 has the largest *P_s* value up to 33.23 μC/cm². The increased *P_s* value plays an important role in improvement of piezoelectric properties.

Raman patterns tested with wave numbers from 100 to 900 cm^{–1} are given in Figure S4a–c₆ to explore the atomic vibration. As shown in Figure S4, all samples have similar Raman profiles and are fitted by Lorentz function.²⁸ Figure S4b exhibits the specific Raman spectra fitting results of CBNLG-C40, which is disintegrated into 16 peaks that labeled from ν₁ to ν₁₆ and are in accord with the previous studies.²⁹ The specific Raman shift versus *x* content of CBNLG-C100*x* ceramics is shown in Figure S4c₁–c₆. For CBN-based ceramics, there are two types of phonon vibration modes, including a high frequency of more than 200 cm^{–1} concerned with the vibration of NbO₆ octahedrons and low frequency below 200 cm^{–1} relevant to the vibration of Bi or Ca ions or both of them in the perovskite-like layers.³⁰ The vibrating modes ν₁–ν₄ change slightly in a small range, due to the approximate radius, ionic bond length with oxygen, and the arrangement of extra-nuclear

electrons of Ce and Gd ions. Moreover, the mode ν_1 slightly tends to low wave numbers for a stronger strength of Gd–O bond than Ce–O bond, confirming the substitution of Gd with Ce in perovskite-like layers. Moreover, Ce ion doping has an impact on models that the Raman shifts higher than 200 cm^{-1} . The reason for this is that Ce^{4+} substitutes Nb^{5+} in perovskite layers, because the ionic radius ($0.87\text{ \AA}$, CN = 6) of Ce^{4+} is close to that of Nb^{5+} ($0.64\text{ \AA}$, CN = 6). Therefore, the vibration of Nb–O octahedrons is slightly affected and then has an effect on resistivity.

To explore the effect of doping on microstructural morphologies, the SEM images of natural surface of CBNLG-C100x ceramics sintered at 1130°C and their average grain size are given in Figure S5a–e. All samples show plate-like grains with random orientation. The growth rate in the a – b plane is higher than the c axis, which is a typical feature of BLSFs. Moreover, some large grains are observed in all samples, due to the existence of regional liquid phase during the sintering process, which increase the sintering driving force.³¹ In addition, no obvious holes and cracks emerge, suggesting a dense microstructure, identified by the high relative density ($>98\%$) calculated with the Archimedes method. The diffusion ability of materials at high temperature can be increased, and thus the sintering activity is enhanced leading to the increased density of the ceramics. As exhibited in Figure S5f, the average grain sizes decrease with increasing x values, which may come from the synergistic effect of oxygen vacancy and the lattice strain energy.³² For the former one, the oxygen vacancy concentration increases with increasing Ce ion, which can promote the growth of grain size along the a – b plane. For the latter one, the smaller ions Nb^{5+} ($0.64\text{ \AA}$, CN = 6) are replaced by larger ions Ce^{4+} ($0.87\text{ \AA}$, CN = 6), which will increase the lattice strain energy,³² thus inhibiting the grain boundary mobility. Under the synergistic effect of the previous two factors, the grain boundary energy occupies the dominant role. Therefore, the grain size decreases with Ce-doping.

For analyzing the chemical state of elements of CBNLG-C100x ceramics, the XPS is measured and exhibited in Figure 3A–E. Cls binding energy (284.60 eV) is used for full spectrum charged correction during the analyzing process. The elements with relatively large quantity, including Bi, Ca, Nb, and O, can be obviously found in Figure 3A. In addition, the weak peak with the binding energy (860 – 920 eV), as shown in Figure 3B, corresponds to the Ce element.³³ The XPS spectrum of Ce3d is fitted with eight peaks labeled ν , ν' , ν'' , ν''' , μ , μ' , μ'' , and μ''' , which indicates the coexistence of Ce^{3+} and Ce^{4+} . In addition, peaks labeled ν , ν'' and μ , μ'' at 880.37 , 885.93 , 895.43 , and 903.57 eV correspond to the final state of Ce^{3+} , and the rest of peaks at 882.49 , 890.04 , 900.22 , and 906.90 eV are related to final state Ce^{4+} . The high resolution XPS

spectrum of Bi4f is given in Figure 3C and fitted into two peaks in CBNLG-C100x ($x = 0.0, 0.2$) ceramics. The peaks A and B are associated with $\text{Bi}4f_{7/2}$ and $\text{Bi}4f_{5/2}$ respectively, revealing the state of Bi^{3+} . The binding energy of Bi4f all shifts to a low value, illustrating the coordination environment change of Bi ions. From Figure 3D, the Nb3d peaks also remove to a high binding energy value, and the reason for this is Ce^{4+} replacing Nb^{5+} , changing the environment of the Nb ion. Figure 3E shows a high-resolution XPS spectrum of O1s, and the peaks at ~ 529.7 and 531 eV are assigned to intrinsic O^{2-} and defect oxygen, respectively. The percentage of intrinsic O^{2-} and defect oxygen is $70.92:29.08$, $68.03:31.97$, when $x = 0.0, 0.4$, respectively. It can be concluded that the oxygen vacancy concentration changes, which is fitted with the tendency of resistance and discussed in the later section.

Z^* plots of CBNLG-0C and CBNLG-40C ceramics in the temperature range from 500 to 700°C are shown in Figure 4A₁,A₂. A parallel circuit model consisting of a resistance (R) and a constant phase angle element is used to fit the test results. The single Cole–Cole semicircle shows that conduction properties are basically dominated by interior grain.¹⁸ Moreover, the center of semicircles is under the x axis, implying that non-Debye relaxation behavior exists in conduction process.³⁴ The semicircular diameter decreases with increasing temperature, meaning a negative temperature coefficient of CBN-based ceramics, which is consistent with the thermally excited ionic conduction mechanism within grains. Meanwhile, at the same temperature, the diameter of Cole–Cole circles changes smaller with varied x values due to the reduction of resistance. For further studying the transmission mechanism of CBNLG-C100x ceramics, the relationship between electric modulus $M''/M''_{\max}$ and frequency of CBNLG-0C and CBNLG-40C ceramics is exhibited in Figure 4B₁,B₂. Theoretically, the relaxation process can be described in a frequency domain, and the formula is as follows³⁵:

$$\frac{X''(\omega)}{X''_{\max}} = \frac{1}{1 - \beta + \frac{\beta}{1+\beta} [\beta (\omega_{\max}/\omega) + (\omega/\omega_{\max})^\beta]}, \quad (2)$$

where $X''(\omega)$ is the imaginary part of frequency-dependent complex modulus, impedance, or dielectric constant, $X''_{\max}$ and $\omega_{\max}$ are the imaginary part maximums and corresponding frequency respectively, and β is the extensibility factor representing the degree of deviating from the ideal Debye model. By fitting the modulus $M''/M''_{\max}$ with formula (2), it can be found that the values of β (0.81 – 0.85) of CBNLG-0C and CBNLG-40C ceramics are both smaller than 1.0 , indicating the relaxation process of deviating from the ideal Debye model. With the increase of temperature, the $M''/M''_{\max}$ peak moves to a higher frequency,

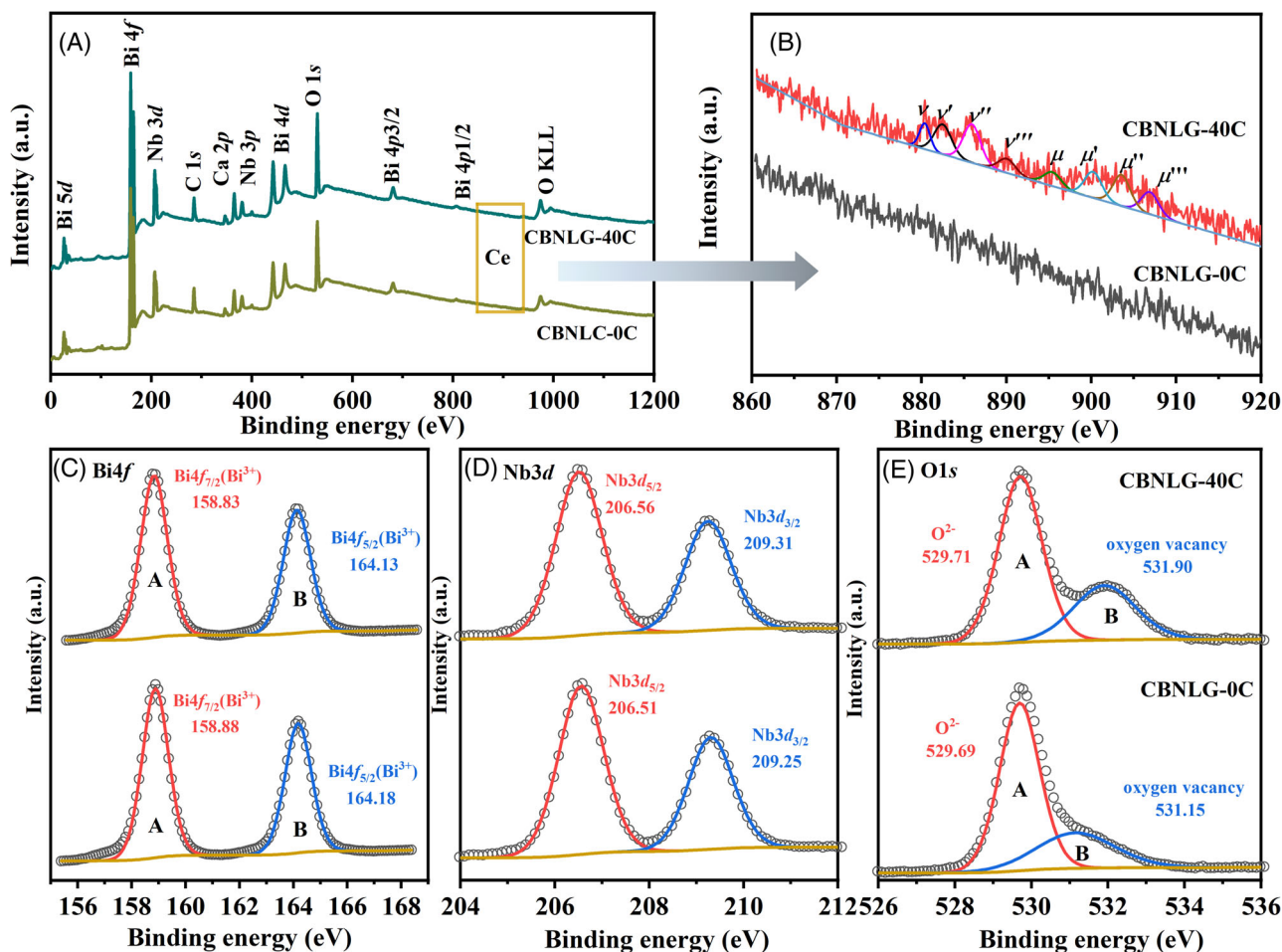


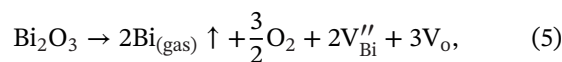
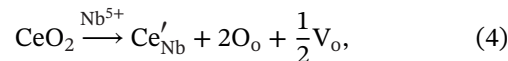
FIGURE 3 X-ray photoelectron spectroscopy (XPS) spectra of CBNLG-C-100x ceramics: (A) $x = 0.00, 0.04$, (B–E) Ce3d, Bi4f, Nb3d, and O1s

indicating a temperature-dependent relaxation process. The frequency at the $M''/M''_{\max}$ peak, $f_{\max}$, which is used to distinguish the long-range effect from the short-range effect, is related to the relaxation activation energy (E_{relax}), and their relationship conforms to Arrhenius law³⁵:

$$f_{\max} = f_0 \exp\left(-\frac{E_{\text{relax}}}{k_B T}\right), \quad (3)$$

where k_B is the Boltzmann constant, T is absolute temperature, and f_0 is the frequency of attempted transition. By fitting the $f_{\max}$ values, the E_{relax} values of CBNLG-0C and CBNLG-40C are shown in Figure 4C, which are 1.80 and 1.23 eV, respectively. Except for CBNLG-0C, the E_{relax} values of other samples are around 1.0 eV, indicating that the migration of oxygen vacancies at high temperature plays a decisive role in dielectric relaxation process. With the x content increasing, the E_{relax} values decrease, which illustrates that Ce ion doping brings more oxygen vacancies. Therefore, the energy of oxygen vacancies diffusion decreases, leading to the smaller E_{relax} values.

In order to analyze the electric resistivity properties, the d_c electric resistivities (ρ) of CBNLG-C100x ceramics are depicted in Figure 4D. The resistivities of all samples decrease with increasing temperature, because of an increasing content of intrinsic carriers under high temperature. The introduction of Ce ions lowers the resistivity but maintains higher than $1 \times 10^5 \Omega \text{ cm}$ at 600°C ($x \leq 0.4$), because of the vitalization of Bi and the substitution of Ce^{4+} to Nb^{5+} . This can be expressed by the Kroger-Vink formula:



Obviously, the previous processes will generate oxygen vacancies, leading to lower resistivity. Moreover, turning points appear at $\sim 425^\circ \text{C}$, implying two conduction mechanisms corresponding to high- and low-temperature region. At the transition temperature, the $\ln \rho$ is in a

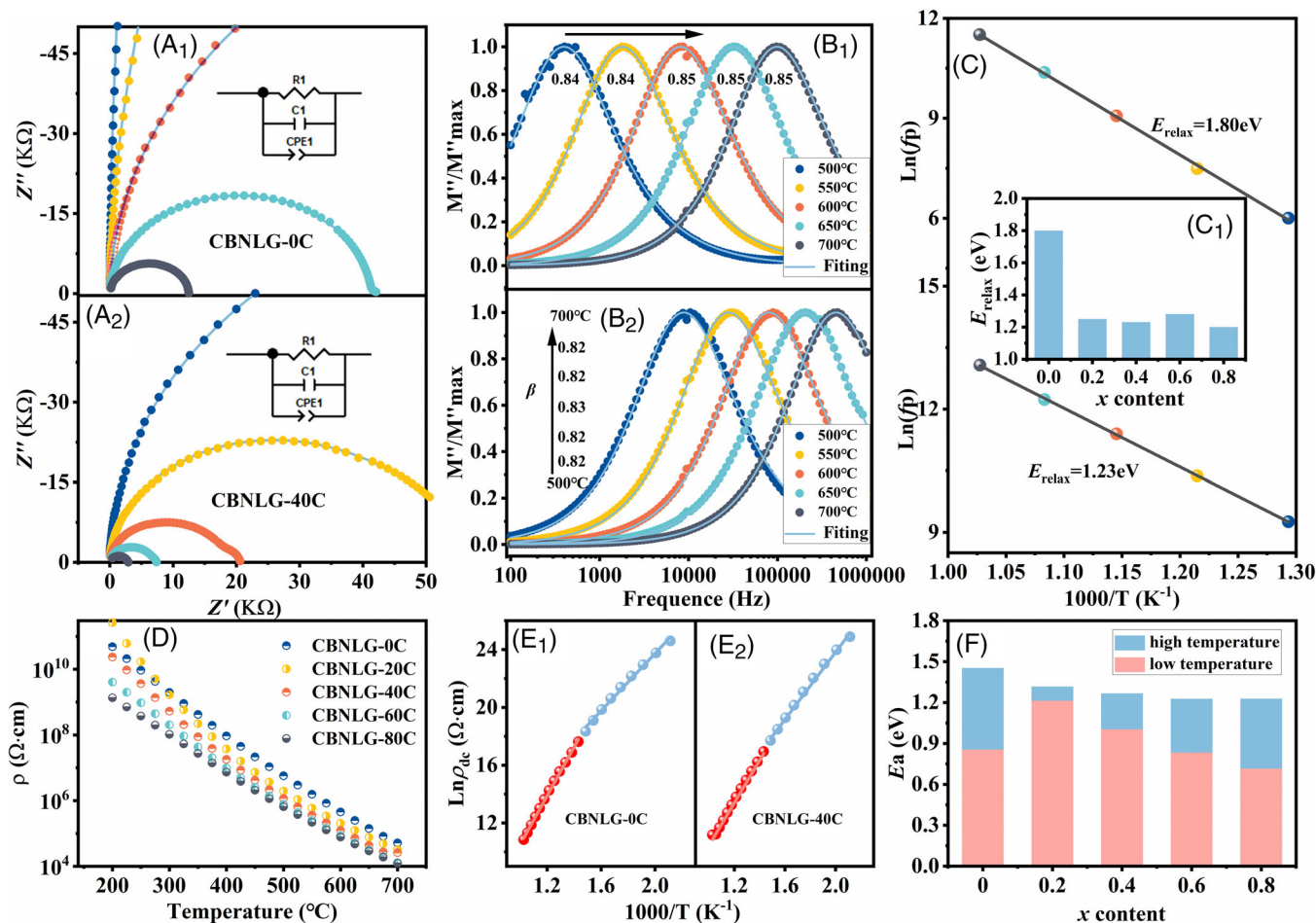


FIGURE 4 (A₁ and A₂) The Z'' plots of CBNLG-0C and CBNLG-40C ceramics in the temperature range from 500 to 700°C; (B₁ and B₂) electrical modulus $M''/M''_{\max}$ versus frequency of CBNLG-0C and CBNLG-40C ceramics; (C) the E_{relax} values of CBNLG-0C and CBNLG-40C ceramics; (C₁) the E_{relax} values of all samples; (D) the ρ_{dc} versus temperature of all ceramics; (E₁ and E₂) the Arrhenius relationship between ρ_{dc} and temperature of CBNLG-C100 x ceramics; (F) the E_a values of ceramics versus x content at two temperature regions

straight line with $1000/T$, which satisfies the Arrhenius relation³⁶:

$$\rho_{\text{dc}} = \rho_0 \exp\left(-\frac{E_a}{k_B T}\right) \quad (6)$$

where ρ_0 is pre-exponential factor, k_B is the Boltzmann constant, T is the absolute temperature, and E_a is the activation energy of charge carriers. As shown in Figure 4E₁, E₂, the E_a is obtained by fitting the experimental data. In general, conduction mechanisms include intrinsic and non-intrinsic conductance. Based on activation energy, the conduction mechanism can be clarified into three parts. The E_a value varying from 0 to 1.1 eV corresponds to the extrinsic conduction,³⁷ varying from 1.1 to 1.4 eV corresponds to intrinsic and non-intrinsic conductance,³⁸ and over 1.4 eV corresponds to the intrinsic conduction.³⁸ It can be seen that the E_a values of CBNLG-C100 x ($x \neq 0$) are about 1.2–1.4 eV in the

high-temperature region, indicating that the conductivity mechanism depends both on the intrinsic and non-intrinsic conductance, and the E_a of CBNLG-0C ceramic is 1.45 eV corresponding to intrinsic conductance.³⁸ In the low-temperature region, the E_a increases first and then declines and reaches its maximum at CBNLG-20C, corresponding with a largest resistivity of CBNLG-20C at a low-temperature region. The E_a values of CBNLG-C100 x ($x \neq 0.2$) is about 0.7–1.0 eV, implying the carriers coming from ionization of oxygen vacancies.³⁷ When $x = 0.2$, the E_a value is 1.2 eV, indicating that the conductivity mechanism depends both on the intrinsic and non-intrinsic conductance.³⁸

The temperature dependences of dielectric constant (ϵ_r) and dielectric loss ($\tan \delta$) of CBNLG-C100 x ceramics at 1 MHz are shown in Figure S6a,b. Dielectric constant of main peak decreases sharply near T_C . Except for the main dielectric peak, an abnormal peak appears when temperature is near to 800°C due to a phase transition, or

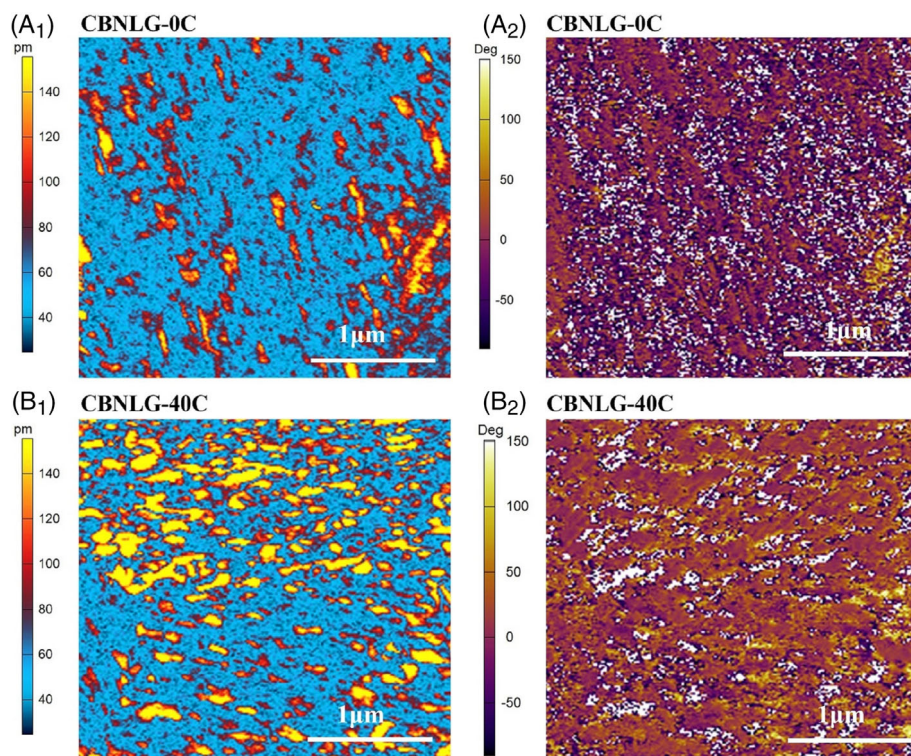


FIGURE 5 (A_1 and B_1) amplitude and (A_2 and B_2) phase images of piezoresponse force microscopy (PFM) for pure CBN and CBNLG-40C with a scanning region of $3 \times 3 \mu\text{m}^2$, respectively

ion-jump relaxation, or space-charge relaxation.³⁹ Besides, from Figure S6b, the $\tan \delta$ values of ceramics keep a considerably low value for a broad temperature range from 25 to 500°C and are even smaller than 4% when temperature equals 500°C. The dielectric loss increases greatly when the temperature increases, which is due to the thermal activation of the intrinsic defect carrier and thus increases the turnover of the defect dipole pair under the external electric field at high temperature.

A domain structure is closely connected with piezoelectric properties.⁴⁰ By using vertical piezoresponse force microscopy (V-PFM), the domain structures of CBNLG-0C ($x = 0.0$) and CBNLG-40C ($x = 0.4$) ceramics are performed in Figure 5A₁, A₂ and B₁, B₂. It can be observed that the substitution of Ce ion decreases the size of domains, thus leading more domains. The built-in electric field enhances with increasing domain-wall density and is accompanied by the improvement of piezoelectric constant.⁴⁰

4 | CONCLUSIONS

In summary, $\text{Ca}_{0.86}(\text{LiGd}_{1-x}\text{Ce}_x)_{0.07}\text{Bi}_2\text{Nb}_2\text{O}_9$ (CBNLG-C100x, $x = 0.0$ – 0.8) ceramics are synthesized by a conventional solid-state reaction method. The structures and domain structures are investigated of CBNLG-C100x ceramics. The crystal structure changes and PFM analysis

show that Ce ion causes smaller and more domains, both of which play an important role in enhancing piezoelectric properties. XPS analysis indicates the coexistence of Ce^{4+} and Ce^{3+} , and the Raman spectra reveal the possibility of Ce^{4+} replacing Nb^{5+} . Therefore, the oxygen vacancy concentration increases with the increasing Ce content, leading to decreased resistivity and enhanced dielectric loss. The $\text{Ca}_{0.86}(\text{LiGd}_{0.6}\text{Ce}_{0.4})_{0.07}\text{Bi}_2\text{Nb}_2\text{O}_9$ ceramic shows good comprehensive performance with high d_{33} (18.0 pC/N), T_C ($\sim 930^\circ\text{C}$), and ρ_{dc} ($1.17 \times 10^6 \Omega \text{ cm}$ at 500°C). In short, $\text{Ca}_{0.86}(\text{LiGd}_{0.6}\text{Ce}_{0.4})_{0.07}\text{Bi}_2\text{Nb}_2\text{O}_9$ ceramic has potential to apply in high-temperature environment.

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CONFLICT OF INTEREST

The authors declare that there is no conflict of interest that could be perceived as prejudicing the impartiality of the research reported.

ORCID

Jingwen Xi  <https://orcid.org/0000-0003-4355-9261>Shangyi Guan  <https://orcid.org/0000-0003-1604-5812>Jie Xing  <https://orcid.org/0000-0003-2800-5519>Jianguo Zhu  <https://orcid.org/0000-0002-2736-9188>

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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